

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

"Black-box APIs provide no guarantee the advertised model is actually being served."

Economic Incentives Drive Covert Model Swaps

Hypothesis test: H_0 (P_{spec}) vs H_1 (P_{actual}) — Can an auditor distinguish the specified model from a substitute?



Smaller Model

Serve 8B parameters
instead of advertised 70B

Massive GPU cost savings
with noticeable quality drop

HIGH RISK



Quantized Variant

FP8/INT8 instead of FP16
precision reduction

Minimal benchmark impact;
nearly undetectable

STEALTHY



Fine-tuned Version

Updated or modified
weights without disclosure

Behavior changes may be
subtle and task-specific

SUBTLE



Different Family

Entirely different model
architecture served

Detectable if large gap,
but can be masked

VARIABLE



Why providers substitute:

GPU costs are the dominant expense in LLM inference. Quantization alone can halve memory requirements, enabling twice the throughput on the same hardware. Black-box APIs provide no mechanism for users to verify what model is actually running.

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Most FP8 score drops fall within error bars of the original model, especially on MMLU and GSM8K — detection is statistically impractical.

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

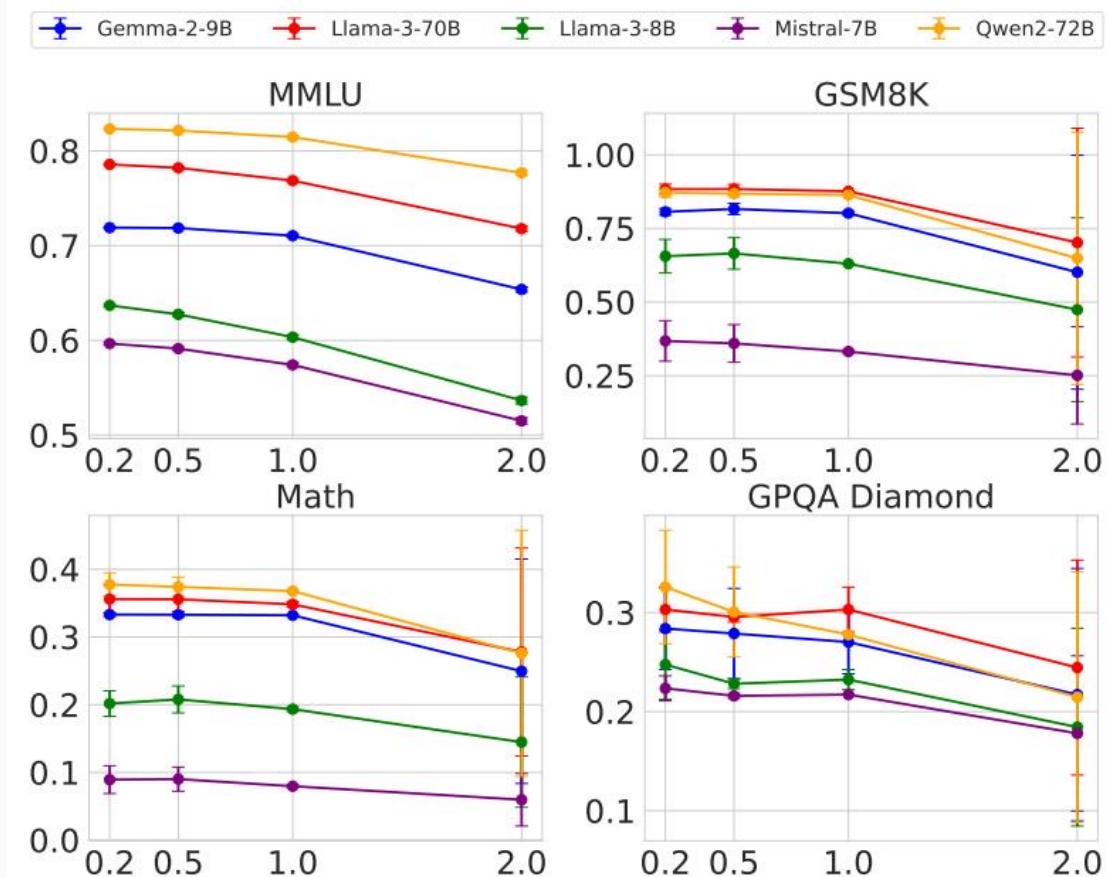
Original precision FP8 quantized Notable drop FP8 higher than original

 **Key insight:** For 4 of 5 model families, MMLU and GSM8K differences are within ±1 standard deviation — statistically indistinguishable. FP8 reduces GPU memory by ~50%, enabling 2x throughput — strong economic incentive with minimal detectable impact.

Source: arxiv.org/abs/2504.04715, Table 1

Higher Temperature Amplifies Performance Gaps but Increases Noise

Temperature scaling degrades all models, further obscuring the substitution signal — the gap between original and substitute narrows or becomes noisy.



Key Observations

- Temperatures tested: 0.2, 0.5, 1.0, 2.0
- At T=2.0, all models show severe performance degradation
- Higher temperature increases output variance, masking substitution signals
- Error bars widen at high temperatures, making FP8 vs original indistinguishable
- Even at low temperature (T=0.2), MMLU/GSM8K gaps remain minimal

! Temperature scaling further obscures the substitution detection signal.

Figure 1: Benchmark accuracy vs temperature for five models across four tasks (Source: arxiv.org/abs/2504.04715)

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

The same model produces different log probabilities across software versions and GPU hardware — indistinguishable from actual substitution.

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Software Frameworks

transformers 4.40 vs 4.50
vLLM 0.6.2 vs 0.8.2

Different versions yield different
log-prob distributions

GPU Hardware

NVIDIA A100 vs H100
Same model, same weights

Hardware-level floating point
differences alter outputs

Detection Failure

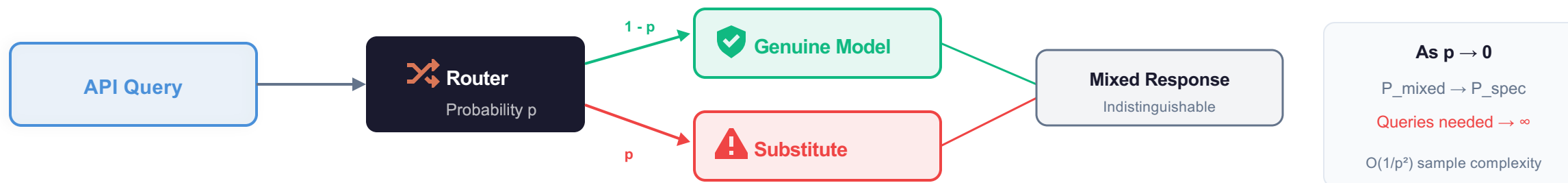
Auditor cannot distinguish
infrastructure variation from
actual model substitution

Log-prob fingerprinting is unreliable

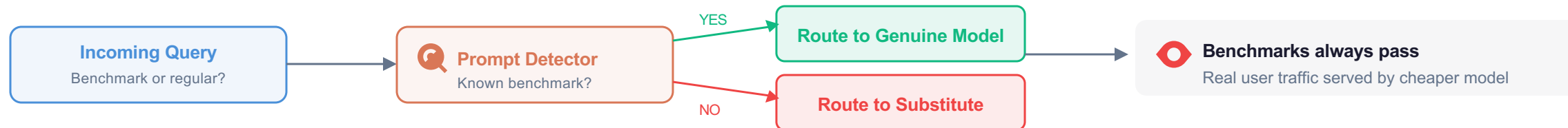
Figure 5: Log probability variation across inference frameworks and hardware for Llama-3-8B (Source: arxiv.org/abs/2504.04715)

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

As the substitution fraction p approaches 0, the mixed distribution converges to the specified model — exponentially more queries needed.



Benchmark Evasion Attack



Combined effect: Randomized substitution + benchmark evasion make software-only detection fundamentally query-prohibitive.
 An adversarial provider can game any audit strategy that relies solely on observing API outputs.

Software-Only Auditing Methods Are Fundamentally Unreliable

All three software-based detection approaches fail under realistic conditions — no software audit can provide reliable substitution guarantees.

Method Type	Core Weakness	Practical Limitation
 Text-Based Statistical Tests e.g., MMD, hypothesis testing on outputs	Requires impractically large query budgets FP8 differences are within noise; statistical power is insufficient	Fails against subtle substitutions (quantization within error bars) Randomized routing at low p makes detection sample-complexity $O(1/p^2)$
 Log-Probability Fingerprinting Token-level log-prob comparison vs reference	Production inference nondeterminism Same model on different GPUs or software versions yields different values	Framework/hardware variation is indistinguishable from substitution Auditor has no baseline: legitimate infra changes look like model swaps
 Adversarial Countermeasures Benchmark evasion + randomized routing	Provider actively circumvents detection Known benchmarks routed to genuine; regular traffic to substitute	Any software audit can be gamed by a motivated provider Fundamental limitation: auditor observes only outputs, never the computation

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

TEEs offer hardware-level attestation: cryptographic proof that the correct model weights are loaded and the correct inference code is running.



Provable

Cryptographic attestation proves specific model weights are loaded in the enclave



Efficient

Modest performance overhead compared to ZKPs which are computationally prohibitive



Robust

Immune to adversarial countermeasures; verification is hardware-enforced, not statistical

Approach	Guarantee	Overhead	Deployable Today
Statistical Tests	None (gameable)	Low	Yes
Log-Prob Fingerprinting	None (nondeterministic)	Low	Yes
Zero-Knowledge Proofs	Cryptographic	Prohibitive	No (for large LLMs)
TEEs	Cryptographic	Modest	Yes

Practical Deployment Path

- NVIDIA Confidential Computing on H100 GPUs provides TEE capability for inference workloads
- Hardware-secured enclaves attest both model weights and inference code at runtime

 Unlike software methods, TEEs do not rely on statistical inference. They provide deterministic, cryptographic proof that the correct computation occurred — no query budget needed.

Close the Verification Gap with Hardware Attestation

1

Model substitution is economically motivated and practically undetectable via software

Providers face strong incentives to reduce GPU costs; black-box APIs offer no verification mechanism.

2

FP8 quantization barely affects benchmarks — the ideal covert substitution

Score drops fall within error bars for most model families on MMLU and GSM8K; halves GPU memory.

3

Inference nondeterminism kills log-probability fingerprinting

Software version and GPU hardware differences produce log-prob variations indistinguishable from substitution.

4

Randomized substitution and benchmark evasion defeat all software audits

Adversarial providers can game any detection method that relies solely on observing API outputs.

5

TEEs are the only currently viable path to verifiable LLM APIs

NVIDIA confidential computing on H100 GPUs enables cryptographic attestation with modest overhead — deployable today.

Demand hardware attestation for your LLM API providers.